

BANARAS HINDU UNIVERSITY

Under Graduate Semester III Examination 2025-26 (NEP SEC)

SEC — CHEMISTRY**Paper No. CHESE31 : Synthetic Methodologies**

Time: 2:30 Hours

Maximum Marks: 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

Note: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

1. Answer all of the following:

- (a) What are the essential characteristics that define a 'click' reaction? [2]
- (b) Write the products for the Ru-catalyzed azide-alkyne cycloaddition reaction. [2]
- (c) Write the product for the reductive amination reaction of ketone using NaCNBH₃. [2]
- (d) Write the names of missing reagents for selective functional group transformations. [2]
- (e) Why is the Grignard reagent prepared in an ether solvent? [2]
- (f) What is the Umpolung reaction? Explain it with suitable examples. [2]
- (g) Despite being highly hydrophobic, why do benzene or toluene still contain a substantial amount of dissolved water? [2]

2. Answer the following:

- (a) Describe the Copper-Catalyzed Azide-Alkyne Cycloaddition (CuAAC) reaction. Why is CuAAC considered the most widely used click reaction? [7]
- (b) Write the thiol-ene click mechanism and explain its applications. [7]

3. Answer the following:

- (a) What factors govern regioselectivity in Ruthenium-Catalyzed Azide-Alkyne Cycloaddition reactions? [7]
- (b) Define bio-orthogonal click chemistry and explain why it is important in chemical biology. [7]

4. Answer the following:

- (a) Write the differences between NaBH₄ and NaCNBH₃ in terms of their reducing abilities and properties. Why is NaCNBH₃ a less powerful reducing agent than NaBH₄? [6]

- (b) Explain the organocatalyzed SuFEx click reaction and describe the mechanism in detail. [4]
 - (c) Write the mechanism for the reduction of an ester and a nitrile using LiAlH_4 . [4]
5. Answer the following:
- (a) Write the product for the amino acid condensation reaction and explain the mechanism. [5]
 - (b) How would you describe the Schlenk equilibrium? [4]
 - (c) Describe how diphenylacetylene can be converted into the corresponding E- and Z-alkenes. Explain the conditions and the underlying reasoning for each transformation. [5]
6. Answer the following:
- (a) Write short notes on liquid chromatography and adsorption chromatography. [5]
 - (b) Write the mechanism for the reaction of carbonyl compounds with Grignard reagents, including the key transition states. [5]
 - (c) Write the products with appropriate mechanism for: (i) Reaction of di-isopropyl ketone with isopropylmagnesium bromide, (ii) Reaction of excess benzaldehyde with phenylmagnesium bromide. [4]
7. Answer the following:
- (a) Describe the origin of the blue color that appears when sodium or potassium metal dissolves in liquid ammonia. How does the electrical conductivity of the solution change with increasing amounts of dissolved metal? [5]
 - (b) Write the mechanism of the Birch reduction and explain each step. What products are formed when naphthalene and anthracene undergo Birch reduction? [5]
 - (c) Describe appropriate drying strategies for the following solvents: acetone, acetonitrile, diethyl ether, and N,N-dimethylformamide (DMF). [4]